

Techniques, Patterns & Best Practices

Every LLM call is a function. Its parameters are English. Senior engineers version, test, and review prompts the same way they version migrations. Today we codify that craft.

How many of your prompts are in git right now?

WHAT'S INSIDE

- 01** The 6-part prompt anatomy
- 02** Three patterns every engineer uses
- 03** XML tags + delimiters
- 04** Self-consistency + reflection
- 05** Measure or don't ship

WHY THIS LESSON MATTERS

Prompts are code. Review them like code.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Zero-shot, few-shot, chain-of-thought patterns.

2

Write a production system prompt.

3

Use delimiters + XML tags for reliable structure.

4

Self-consistency + reflection for hard reasoning.

5

Build a mini eval harness.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

The 6-part prompt anatomy

Role → Context → Task → Examples
→ Input → Output format

02

Three patterns every...

Key concept of this lesson.

03

XML tags + delimiters

The reliability trick

04

Self-consistency +...

When zero-shot fails

01

PART 1 OF 4

The 6-part prompt anatomy

Role → Context → Task → Examples → Input → Output format

CONCEPT 1 OF 4

The 6-part prompt anatomy

ROLE — "You are a senior Indian tax consultant."

CONTEXT — domain rules, constraints.

TASK — specific instruction with exact output shape.

EXAMPLES — 1-3 input/output pairs.

INPUT — the current request, clearly delimited.

OUTPUT FORMAT — JSON schema or strict structure.

MEMORISE

6 parts

Skip any one → unreliable output.

DEFINITION

The 6-part prompt anatomy

Role → Context → Task → Examples → Input → Output
format

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

Three patterns every engineer uses

A core idea you'll use throughout this lesson.

KEY POINTS

- A core idea you'll use throughout this lesson
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

CONCEPT 3 OF 4

XML tags + delimiters

Wrap input: <input>...</input>.

Wrap examples: <example><user>...</user><output>...</output></example>.

Wrap sections: <context>...</context>.

Models trained on XML respect tags far better than paragraph text.

RULE

Tag everything

5 minutes of XML = hours saved on debugging.

DEFINITION

XML tags + delimiters

The reliability trick

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Self-consistency + reflection

Self-consistency: sample N times at temp>0, majority vote.

Reflection: "Critique your answer and revise."

Both add cost; both add accuracy on hard reasoning.

Measure before enabling in production.

USE FOR

Hard reasoning

Don't apply to simple extraction.

KEY POINTS

- Self-consistency: sample N times at temp>0, majority vote
- Reflection: "Critique your answer and revise."
- Both add cost; both add accuracy on hard reasoning
- Measure before enabling in production

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Upgrade a bad prompt

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

- 1
- 2 Rewrite using the 6-part anatomy + XML tags.
- 3 Run both on the same input; compare output.
- 4 Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Which prompt in your codebase is most likely to fail silently?

Q2

Do you have a golden set for any LLM feature?

Q3

What would it take to bring prompts under version control this week?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Six parts of a production prompt:

A

Name, Age, City, ...

B

Role, Context, Task, Examples, Input, Output format

C

Open, Write, Close

D

Just the question

Six parts of a production prompt:

CORRECT ANSWER



Role, Context, Task, Examples, Input, Output format

WHY

Memorise these six — missing any one leads to unreliable outputs

Chain-of-thought is used for:

A

Simple classify

B

Multi-step reasoning

C

Speed

D

Cost saving

Chain-of-thought is used for:



CORRECT ANSWER

Multi-step reasoning

WHY

"Let's think step by step" improves math and logic tasks notably

Why XML tags?

A

Prettier

B

Models respect structure better

C

Smaller

D

Faster

Why XML tags?



CORRECT ANSWER

Models respect structure better

WHY

LLMs trained on tagged data follow delimited structure much more reliably

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

ROLE — "You are a senior Indian tax consultant."

CONTEXT — domain rules, constraints.

TASK — specific instruction with exact output shape.

EXAMPLES — 1-3 input/output pairs.

INPUT — the current request, clearly delimited.

OUTPUT FORMAT — JSON schema or strict structure.

MEMORISE

6 parts

Skip any one → unreliable output.

THE TAKEAWAY

"Review. Test. Version. Treat them that way."

WHAT TO NOTICE

1

The 6-part prompt anatomy

2

Three patterns every...

3

XML tags + delimiters

4

Self-consistency +...

Mini assignment — build a golden set

Pick a task you care about. Build a 20-case golden set + 3 prompt versions.

DELIVERABLES

- 20 inputs with expected outputs.
- Zero-shot, few-shot, CoT versions.
- Score each. Record cost.
- Post the winner + why.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Spreadsheet with accuracy per version.

ONE THING TO REMEMBER

“

"Review. Test. Version. Treat them that way."

— AI Learning Hub

END OF LESSON 5

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 3 — RAG. Give your LLM a memory.